HIGH-PERFORMANCE NATIVE WINDOWS FLIGHT SIMULATOR

System Specification & Domain Skill | Target: C++20/C++23 & Direct3D 12 | Architecture: YSFlight-Class Efficiency Baseline | Validation GPU: AMD Radeon RX 6700 XT (RDNA 2)

1. Diagnóstico Técnico & Delimitação de Restrições

Deconstruction & Context Analysis:

- **Source Target:** `prompt_gcollab.txt.pdf` (40 pages, 38 engineering sections).
- **Core Mission:** Lead architect & systems developer for a clean-slate, native Windows flight simulation platform that rejects heavyweight general-purpose game engines (Unreal, Unity, Godot) in favor of high resource efficiency, sub-second boot times, consistent frame pacing, and rigorous aerospace dynamics (YSFlight performance philosophy with modern rendering).
- **Runtime Target:** Windows 10/11 x64, Direct3D 12 (native runtime detection, Agility SDK evaluated on demand), primary validation on AMD Radeon RX 6700 XT (RDNA 2, 12GB VRAM).

Architectural Vector	Mandatory Invariant	Strictly Prohibited Anti-Pattern
Workflow Governance	Phase 0 requirements discovery protocol must precede code emissions. Requirements committed to `docs/REQUIREMENTS.md`.	Immediate generation of complete simulators, unverified tool installations, or mock execution claims.
Engine Platform	Pure Win32 C++20/C++23 native systems, Direct3D 12, headless execution capability.	Complete commercial engines (Unreal, Unity, Godot), D3D11 wrappers, unverified heavy middleware.
Threading & Memory	Fixed-step physics accumulator, snapshot-based double/triple buffer synchronization, bounded queues.	Uncontrolled busy-waiting, unbounded job queues, false sharing, premature/unverified lock-free code.
Flight Mechanics	6-DOF rigid-body dynamics, quaternion integration, tabulated aerodynamic coefficients, SI unit core.	Euler-angle singularities, teleporting autopilot controls, mixing imperial/metric without explicit converters.
Graphics Pipeline	Explicit D3D12 resource states, descriptor heaps, camera-relative floating-origin rendering, RDNA 2 profiling.	Mandatory ray tracing, single-precision world Cartesian coordinates causing large-scale vertex jitter.

2. Execução Direta da Solução (Código / Arquitetura / Parâmetros)

```
<skill name="native-flight-simulator-engine" version="1.0">
  <metadata>
    <role>Principal Engine Architect & Aerospace Simulation Lead</role>
    <target_os>Windows 10 / 11 x64</target_os>
    <graphics_api>Direct3D 12 Native</graphics_api>
    <validation_gpu>AMD Radeon RX 6700 XT (RDNA 2, 12 GB VRAM)</validation_gpu>
    <standard>C++20 baseline (C++23 verified compiler features)</standard>
  </metadata>

  <subsystem_matrix>
    <!-- Tier 0: Platform, Infrastructure & Math -->
    <subsystem id="SimulatorCore" scope="Lifecycle, main loop coordination, state machine" />
    <subsystem id="PlatformSystem" scope="Win32 windowing, event pump, raw OS handles" />
    <subsystem id="TaskScheduler" scope="Worker thread pool, job dependency DAG, fiber/task queue
s" />
    <subsystem id="FileSystem" scope="Async buffered I/O, schema-validated file handles" />
    <subsystem id="AssetSystem" scope="glTF 2.0 parser, cache, streaming lifetime tracker" />
    <subsystem id="ConfigurationSystem" scope="Hierarchical user/engine settings, profile serializ
ation" />
    <subsystem id="DiagnosticsSystem" scope="Overlay telemetry, PIX/RGP instrumentation, leak dete
ction" />

    <!-- Tier 1: Real-Time Aerospace Simulation (Headless Capable) -->
    <subsystem id="FlightDynamicsSystem" scope="6-DOF rigid body, quaternion kinematics, mass/iner
tia tensor" />
    <subsystem id="AerodynamicsSystem" scope="Tabulated CL/CD/CY/Cl/Cm/Cn tables, NACA solver, sta
ll modeling" />
    <subsystem id="PropulsionSystem" scope="Piston, turboprop, turbofan, jet models, fuel consumpt
ion" />
    <subsystem id="GroundDynamicsSystem" scope="Landing gear struts, tire friction, spring-damper
suspension" />
    <subsystem id="CollisionSystem" scope="Spatial hash grid, terrain height queries, continuous c
ollision" />
    <subsystem id="WeatherSystem" scope="ISA atmospheric model, pressure altitude, turbulence, win
d shear" />

    <!-- Tier 2: Avionics, Systems & Hardware I/O -->
    <subsystem id="InputSystem" scope="RawInput/DirectInput/HID/XInput abstraction, calibration, d
eadzones" />
    <subsystem id="SerialInterfaceSystem" scope="Async COM port framing, checksummed binary cockpi
t telemetry" />
    <subsystem id="AvionicsSystem" scope="Six-pack flight instruments, sensor lag, bus electrical
state" />
    <subsystem id="NavigationSystem" scope="VOR, DME, ILS Localizer/GlideSlope, navaid DB, magneti
c variation" />
    <subsystem id="ScenarioSystem" scope="Spawning, initial conditions, weather presets, flight re
cording" />
    <subsystem id="NetworkingSystem" scope="Client/server UDP/TCP, snapshot replication, dead reck
oning" />
    <subsystem id="TelemetrySystem" scope="Blackbox logging, CSV/binary flight performance recorde
r" />

    <!-- Tier 3: Rendering & Audio Presentation -->
```

```
    <subsystem id="GraphicsSystem" scope="Direct3D 12 renderer, command lists, root signatures, PS
Os" />
    <subsystem id="AudioSystem" scope="XAudio2/WASAPI spatial 3D audio, RPM/airflow sound synthesi
s" />
  </subsystem_matrix>

  <execution_rules>
    <rule id="R01" category="Phase0Gate">
      Under no circumstances emit implementation code before requirements approval.
      Execute Phase 0: Ask focused questions on Environment, C++ standard, Architecture,
      Aircraft category, Flight dynamics fidelity, Graphics tier, Terrain, Network, Hardware I/O,
      and Initial milestone. Record results in docs/REQUIREMENTS.md.
    </rule>
    <rule id="R02" category="ThreadingModel">
      Physics must execute on a dedicated fixed-timestep thread (60Hz / 120Hz / 240Hz).
      Renderer must run asynchronously at variable refresh rate.
      Communication between simulation and rendering must use lock-free or triple-buffered snapsho
ts.
      Never allow variable render dt to enter physics integration.
    </rule>
    <rule id="R03" category="HeadlessCapability">
      Engine architecture must support running complete flight dynamics and avionics without
      initializing Direct3D 12 or audio subsystems.
    </rule>
    <rule id="R04" category="NumericsAndCoordinateFrames">
      Core physics must use SI units internally (meters, kilograms, seconds, radians).
      Aviation units (knots, feet, inHg, lbs) must be isolated to boundary conversion utilities.
      Rotational state must be maintained via unit quaternions. Euler angles are strictly for disp
lay.
      Frames defined: World (ECEF/Local), Navigation (NED), Body (X-fwd, Y-right, Z-down), Wind (a
lpha, beta).
    </rule>
    <rule id="R05" category="FloatingOrigin">
      To avoid 32-bit floating point precision loss across large terrain, rendering must implement
      Camera-Relative / Floating-Origin coordinates (World pos stored in double, shifted before D3
D12 submission).
    </rule>
    <rule id="R06" category="Direct3D12Management">
      DirectX 12 must manage explicit resource states and barriers.
      Descriptor heaps must use frame-budgeted indexing.
      Implement swapchain triple-buffering with synchronization fences.
      All shader pipelines must be precompiled via DXC. Ray tracing is strictly optional/decouple
d.
    </rule>
    <rule id="R07" category="MemoryDiscipline">
      No dynamic memory allocations (new/malloc) in simulation or rendering hot loops.
      Use preallocated memory arenas, frame allocators, and object pools.
      Track process working set, committed memory, and GPU VRAM budget.
    </rule>
    <rule id="R08" category="ValidationDiscipline">
      Never claim successful compilation, tool execution, or framerate performance without
      verified system diagnostics and execution output.
    </rule>
  </execution_rules>
</skill>
```

Milestone Execution Roadmap (M0 — M22)

Milestone	Deliverable Scope	Key Acceptance Criteria
M0	Requirements Discovery & Architectural Baseline	Signed-off `docs/REQUIREMENTS.md` and `docs/ARCHITECTURE.md`.
M1	Toolchain, CMake & Minimal Build	Clean compilation on MSVC with warning level 4/WX.
M2	Native Win32 Application Lifecycle	Window creation, high-DPI handling, message pump, clean exit.
M3	Direct3D 12 Core Initialization	Device creation, command queues, swapchain presentation, clear screen.
M4	GPU Adapter Enumeration & Capability Probe	RDNA 2 RX 6700 XT detection, fallback adapter handling.
M5	Math, Units & Coordinate Systems	Quaternions, SIMD vector math, SI <-> Aviation unit conversion suite.
M6	Fixed-Timestep Simulation Scheduler	Deterministic accumulator loop, sub-step interpolation, pause/resume.
M7	6-DOF Rigid Body Physics Engine	RK4 integration, inertia tensor, gravity, ballistic trajectory tests.
M8	Aerodynamic Coefficient Engine	CL/CD/CY lookup, alpha/beta incidence, lift/drag polar calculation.
M9	Input Engine & Hardware Abstraction	RawInput/DirectInput joystick axis curves, deadzones, keyboard fallback.
M10	First Controllable Flight Prototype	Controlled pitch/roll/yaw, basic flight envelope validation.
M11	Terrain Heightfield & Ground Reactions	Runway collision, oleo-pneumatic strut spring/damper, tire friction.
M12	Propulsion & Fuel Subsystem	Piston/propeller thrust calculation, throttle response, fuel consumption.
M13	glTF 2.0 Aircraft Model Loading	Mesh loading, materials, hierarchical control-surface transforms.
M14	Analog Six-Pack Avionics	IAS, Altimeter, Attitude, Turn Coordinator, Directional Gyro, VSI.
M15	Spatial Audio Subsystem	XAudio2 3D listener, engine RPM audio pitch shifting, wind sound.
M16	Radio & Navigation Instruments	VOR radial reception, DME distance, ILS localizer & glide slope needles.
M17	Serial Cockpit Hardware Interface	Async COM port telemetry stream, microcontroller switch/potentiometer input.
M18	Atmospheric & Weather Systems	ISA atmospheric gradient, wind vector, thermal turbulence injection.

Milestone	Deliverable Scope	Key Acceptance Criteria
M19	Graphics Optimization & Visual FX	Cascaded shadow maps, atmospheric fog, terrain LOD, instanced scenery.
M20	Multiplayer Network Prototype	UDP snapshot replication, entity interpolation, dead reckoning.
M21	Glass Cockpit & Advanced Autopilot	PID heading/altitude hold, flight director, digital flight display.
M22	Full Scenery & Multi-Aircraft Expansion	Streaming geographic terrain tiles, multi-archetype flight models.

3. Blindagem de Casos de Borda & Invariantes Críticos

Failure Mode / Edge Case	Failure Mechanism	Enforced Mitigation / Invariant
Gimbal Lock & Quaternion Drift	Successive integration of angular rates accumulates numerical error, causing non-unit quaternion lengths and orientation skew.	**Renormalization Invariant:** Renormalize quaternion at every integration step: `q = q / |q|`. Assert `||q| - 1.0| < 1e-5`. Use quaternions for state; Euler angles calculated only for UI/gauges.
Floating-Point Jitter at Scale	Single-precision 32-bit floats lose sub-millimeter precision past ~10km from origin, inducing vertex jitter in cockpits and runways.	**Floating-Origin Invariant:** Store global world coordinates in `double` (64-bit). Compute vertex transforms relative to camera position (Camera-Relative Rendering) before feeding D3D12 constant buffers.
Physics Timestep Tunneling	Large frame drops in variable render loop cause massive delta times, causing aircraft landing gear to penetrate terrain geometry.	**Fixed Accumulator Invariant:** Decouple physics from render delta time. Cap accumulator maximum delta to prevent the "spiral of death" (e.g. max 5 sub-steps per frame); execute fixed 120Hz physics sub-steps.
GPU Device Loss (TDR)	Long shader runs or driver reset triggers `DXGI_ERROR_DEVICE_REMOVED`, crashing unhandled native D3D12 applications.	**Device Recovery Protocol:** Query `GetDeviceRemovedReason()`, destroy all device-dependent resources, flush queues, and reinitialize device state without terminating simulator core.
Explosive Ground Penetration	Discrete collision detection detects contact below terrain plane, calculating violent impulse responses that launch the aircraft.	**Spring-Damper Contact Invariant:** Model suspension struts with continuous spring-damper equations with compression limits and restitution clamping. Use continuous swept tests for high-speed contact.
Premature Code Emission	Model generates hundreds of speculative C++ files before agreeing on IDE, build system, or architectural choices.	**Phase 0 Gate:** Intercept any attempt to write simulator files before `docs/REQUIREMENTS.md` is reviewed and confirmed by user.